

# AutoValuate Intelligence

AI-powered used-car valuation for the UAE — free, browser-based, deployed

Executive Brief · 3 pages

SP Jain School of Global Management — Student Venture

27 July 2026

AutoValuate Intelligence is a **live, working web application** that tells a used-car buyer or seller in the UAE what a car is really worth — in seconds, for free. It combines two independent AI systems: a **computer-vision model** that inspects uploaded photos for physical damage, and a **machine-learning pricing model** trained on real UAE listings and vehicle specifications. The product runs entirely on free-tier cloud infrastructure, built and operated by a student team.

**13.18%**

median pricing error — 52% better than a naive make/model baseline

**1,302**

real UAE listings used to train the pricing model

**79%**

of true prices fall inside our stated confidence range (target 80%)

**AED 0**

cost to the user — no paid tiers, no sign-up required

## 01 What the product does

- **Instant price estimate** — enter a car's details, get a price range with a stated confidence level, not a single guessed number.
- **On-device damage scan** — upload photos; an AI inspects them for dents, scratches, cracks and more. Photos never leave the user's device.
- **Explainable pricing** — every estimate shows exactly which factors (mileage, age, brand, specs, damage) moved the price and by how much.
- **Market comparables** — the estimate is shown against real, similar listings currently on the market.
- **Dealer bulk valuation** — a dealer can upload a spreadsheet of an entire fleet and get every car valued at once.
- **Open, free API** — any developer can call the pricing engine directly; no account or key required.

## 02 How the pieces fit together



All four stages run on free-tier cloud services — no paid infrastructure.

## 03 Technology stack, in plain terms

| LAYER        | WHAT WE USE                                               | WHAT IT'S FOR                                                              |
|--------------|-----------------------------------------------------------|----------------------------------------------------------------------------|
| Website      | Next.js (React) on Vercel                                 | The app users see and interact with in their browser                       |
| Damage scan  | YOLOv8 object-detection model, run in-browser             | Finds and labels visible car damage in a photo, on the user's own device   |
| Price engine | XGBoost (gradient-boosted trees)                          | Predicts fair market price from a car's specs and condition                |
| Backend      | FastAPI (Python) on Render, with an AI agent orchestrator | Combines the price engine, the damage scan and market data into one report |
| Database     | Supabase (Postgres + vector search)                       | Stores comparable listings and shareable report links                      |

## 04 How the two AI engines work

**Engine 1 — Damage Detection (Computer Vision)**

A model called **YOLOv8** looks at each uploaded photo and draws a box around anything that looks like damage — a dent, a scratch, a crack, broken glass, a broken light, a flat tire, a puncture, or a missing part (8 categories in total). It converts what it sees into a single condition score out of 100. This runs entirely on the user's own phone or laptop, so no photo is ever sent to our servers — a genuine privacy advantage, not just a technical choice.

**Engine 2 — Price Estimation (Machine Learning)**

A model called **XGBoost** was trained on 1,302 real UAE car listings, enriched with each vehicle's physical specifications (horsepower, torque, fuel economy) joined in from an independent UAE spec database. Given the make, model, age, mileage, specification and condition, it predicts a fair price — and, importantly, a **range** around that price with a stated confidence level, rather than pretending to know an exact number.

**The Orchestrator**

A layer of coordinating logic takes the output of both engines, pulls in comparable listings, and assembles one explainable report — the price, why it is that price, and how confident the system is.

## 05 Key terms explained

**YOLOv8 / ONNX**

YOLOv8 is the damage-detection AI. ONNX is the portable file format it's exported to so it can run inside a web browser instead of on a server.

**XGBoost**

A widely-used, highly accurate machine-learning technique for predicting a number (here, price) from structured data like mileage, age and vehicle specifications.

**mAP (mean Average Precision)**

The standard score for "how good is the damage detector" — the industry way to grade computer-vision accuracy.

**APE / Median APE**

"Absolute Percentage Error" — how far off a single price guess was, in percent. We report the median across all test cars, which is more honest than an average.

**Confidence range (split-conformal prediction)**

A statistically calibrated price range — e.g. "80% confident the true price falls in this band" — rather than one number presented as certain.

**Agentic orchestration**

AI "agents" that each handle one job (pricing, damage, market lookup, write-up) and hand results to each other automatically to build the final report.

## 06 Results — verified, not marketing

| METRIC                          | RESULT         | WHAT IT MEANS                                                                             |
|---------------------------------|----------------|-------------------------------------------------------------------------------------------|
| Pricing accuracy (median error) | <b>13.18%</b>  | Typical price estimate is within ~13% of the actual sale price                            |
| Improvement vs. naive baseline  | <b>+52.3%</b>  | More than twice as accurate as simply averaging similar listings                          |
| Confidence-range reliability    | <b>79%</b>     | Very close to the 80% target — the stated ranges can be trusted                           |
| Vehicle-spec match rate         | <b>81.0%</b>   | Physical specs (hp, torque, etc.) successfully matched for 4 in 5 cars, boosting accuracy |
| Damage-detector accuracy (mAP)  | <b>0.732</b>   | Strong on close-up damage photos; see honesty note below                                  |
| Browser ↔ server scoring match  | <b>56 / 56</b> | The score shown to the user always matches what the backend independently computes        |

**Read this the way a CFO would want it read:** the 0.732 detector score was measured on a validation set that overlaps with training data and covers only 6 of 8 damage types — it is not a number we would defend to an external auditor. The pricing numbers above, by contrast, are measured on a genuinely held-out test set and are safe to repeat externally.

## 07 Known limitations — stated plainly

- 1 The damage scan is sensitive to how a photo is framed.**

The same damage, photographed at a slightly different crop or distance, can score up to 47 points differently. We have diagnosed the root cause precisely (the detector was originally trained on close-up damage photos, not whole-car shots) and scoped the fix — retraining with wide-shot images — but have not yet been able to run it, since it requires more real UAE car photos and GPU training time we don't yet have.
- 2 Pricing accuracy is still capped by data volume, even after a real gain.**

Joining in vehicle specs (hp, torque, fuel economy) dropped median error from 15.65% to 13.18%. The floor is still data-bound: at 1,302 listings it's ~6.9% with this feature set; the ~8–10% figure quoted for larger markets needs ~6,000–14,000 listings. More data, not tuning, is the remaining lever.
- 3 A licensing question blocks closed-source commercial resale.**

The damage detector is derived from an open-source model licensed AGPL-3.0. That license permits free use exactly as we operate today, but would require either a paid commercial license or a replacement model before this could be sold as closed-source software.
- 4 There is currently no revenue.**

The product is intentionally 100% free with no paid tiers — a deliberate choice while the data and licensing questions above remain open, not an oversight.
- 5 Coverage is UAE-only.**

All training data, comparable listings and market assumptions are specific to the UAE market today.

## 08 What's next

- |                                                                                                                                                                     |                                                                                                                                |                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br><b>Retrain the damage detector</b><br>Run the already-written retraining process on a larger, wide-shot photo set to fix the framing sensitivity above. | <b>2</b><br><b>Grow the listings dataset</b><br>Expand past 1,302 UAE listings — the single biggest lever on pricing accuracy. | <b>3</b><br><b>Resolve the licensing path</b><br>Decide between an enterprise license or a licence-compatible replacement model before any commercial launch. |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 09 Project context & approach

### Who built this

A student team at SP Jain School of Global Management, built and operated as a combined academic and venture-track project — not a funded startup.

### Why free-tier only

Every service in the stack (hosting, database, model inference) runs on a free pricing plan by design. It proves the product works before any cash is spent, and keeps the cost of serving a user at effectively zero.

### How claims in this document are checked

Every number above is read from a committed evaluation report, re-runnable from the source repository — nothing here is an estimate or a rounded-up marketing figure.

### How limitations are found

Weak spots (like the framing sensitivity on page 3) are found by deliberately trying to break the system before a user does, not discovered after launch.

**Live product:** [auto-valuate-intelligence.vercel.app](https://auto-valuate-intelligence.vercel.app)  
**API:** [autovaluate-api.onrender.com](https://autovaluate-api.onrender.com)

**Source & full evaluation reports:** [github.com/krish2105/AutoValuate-Intelligence](https://github.com/krish2105/AutoValuate-Intelligence)  
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